

Decision Log

Database Selection

Chosen: SQLite

Options Considered:

- DuckDB (analytical, fast)
- SQLite (simple, reliable)
- PostgreSQL (production-grade)

Rationale:

- No server setup required
- Built into Python standard library
- Works reliably on Windows
- Sufficient for all required queries

Trade-offs:

- Less analytical functions than DuckDB (e.g., PERCENTILE_CONT)
- Slower for very large datasets
- But acceptable for 100K+ row datasets

Implementation: conn =

```
sqlite3.connect('data/processed/nyc/nyc.db')
```

```
df = pd.read_sql_query(query, conn)
```